

Medicinal uses

Pigments don't just give living things their color. They can also do other important jobs. Chika used the pigment quercetin—extracted from onion skin—to invent a medicine to help people with high blood pressure.

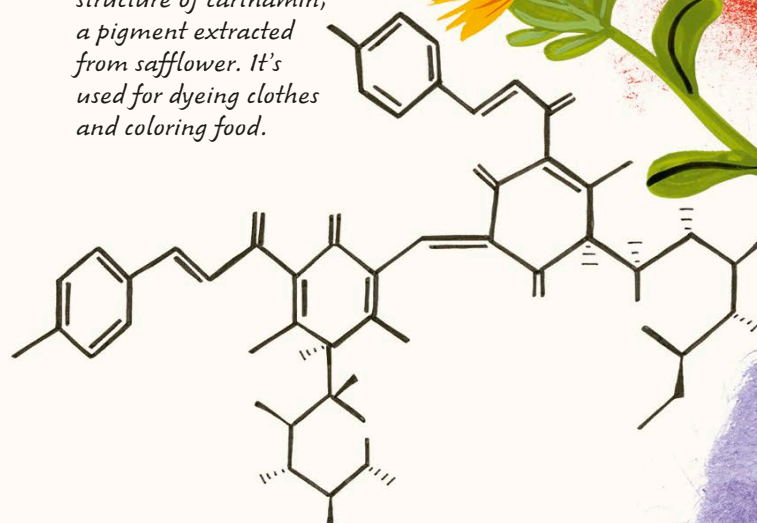


Colorful chemicals

Chika wanted to research natural pigments—the chemicals that give plants and animals their rainbow colors. Her first big project focused on a plant called purple gromwell. She learned how to extract (take out) pigments from the plant's roots, making crystals of each colorful chemical.

Then, Chika investigated the structure of these crystals to figure out what they were made from. After naming the purple gromwell's pigment shikonin, she moved to look at another red pigment called carthamin, which was taken from safflowers to be used in dyes. It took five years of hard work for Chika to discover carthamin's chemical structure. This breakthrough helped her become the second woman in Japan to earn a doctorate—the highest level of degree you can get!

This is the chemical structure of carthamin, a pigment extracted from safflower. It's used for dyeing clothes and coloring food.



Chika also found out about pigments from other plants, such as eggplants and black beans.

Black beans



Eggplant

